

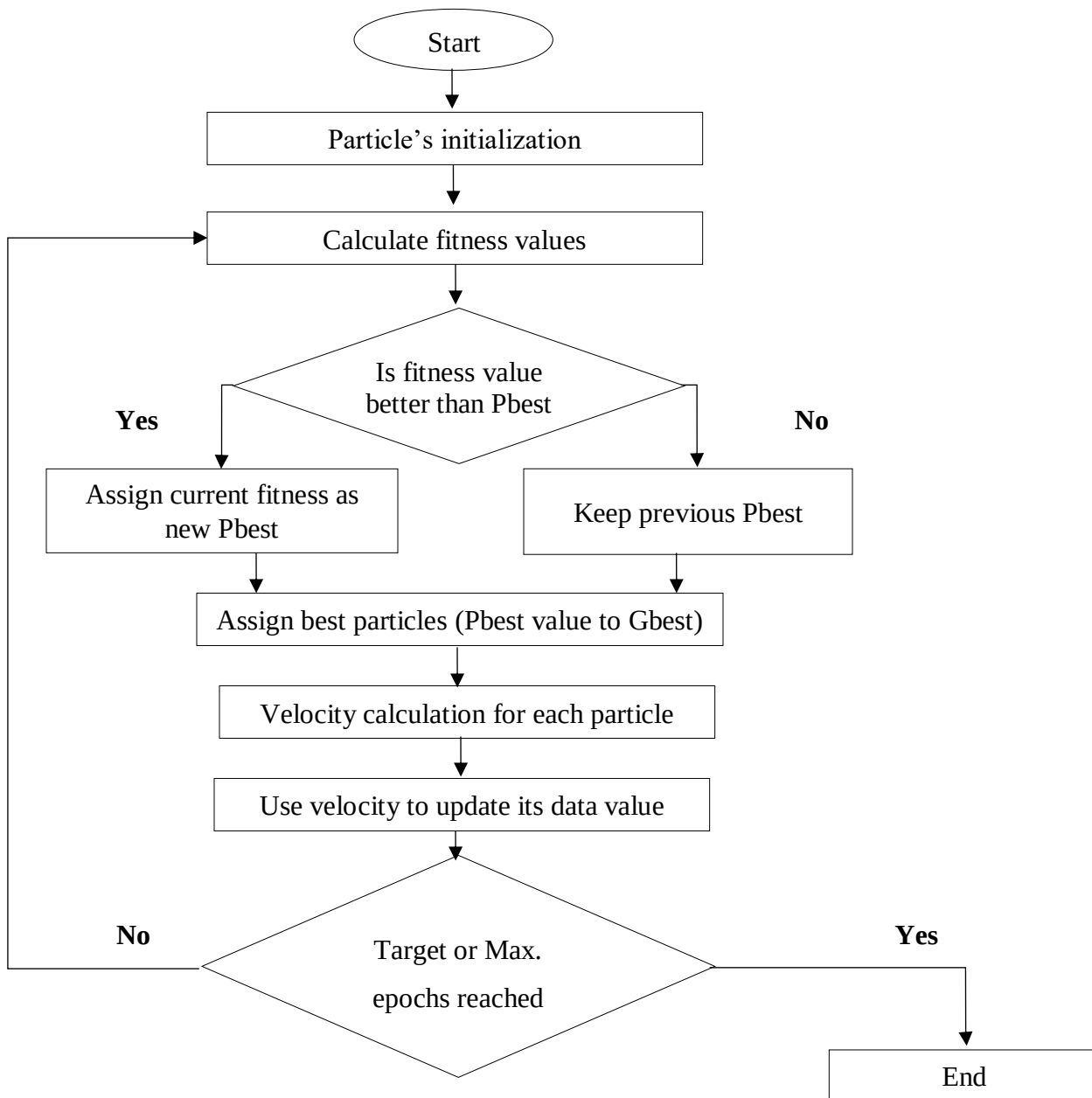


Figure 3.7 Flowchart of Basic PSO

The performance of PSO is dependent on three control parameters which are inertial weight (w), Acceleration Coefficient ($C1$ & $C2$), and Random number ($R1$ & $R2$). Inertial weight is used for maintaining the effect of convergence and diversity. A large value of w indicates better global exploration, while the small value of w works on exploitation. Unbalanced values containing the parameters can hurt the results, such that if we take the value of $c1$ and $c2$ is low, it tends to a smooth particle trajectory and abrupt movements in case of high values of $c1$ and $c2$. Similarly, if $c1$ is much greater than $c2$, it tends to excessive wandering and causes premature convergence

when c_1 is much less than c_2 . Large inertial weight greater global search ability and smaller inertial weight greater local searchability. By dynamic changing of Inertial weight (w), the acceleration coefficient (c_1 & c_2), efficiently explores the search space.

PSO is a robust stochastic optimization technique based on the movement and intelligence of swarms. PSO applies the concept of social interaction to problem solving. It was developed in 1995 by James Kennedy (social-psychologist) and Russell Eberhart (electrical engineer). It uses a number of agents (particles) that constitute a swarm moving around in the search space looking for the best solution. Each particle is treated as a point in a N dimensional space which adjusts its “flying” according to its own flying experience as well as the flying experience of other particles. Each particle keeps track of its coordinates in the solution space which are associated with the best solution (fitness) that has achieved so far by that particle. This value is called personal best, $pbest$.

Another best value that is tracked by the PSO is the best value obtained so far by any particle in the neighborhood of that particle. This value is called $gbest$.

The basic concept of PSO lies in accelerating each particle toward its $pbest$ and the $gbest$ locations, with a random weighted acceleration at each time.

Velocity

$$V_i^{t+1} = wV_i^t + c_1r_1(PBest_i - X_i^t) + c_2r_2(GBest - X_i^t)$$

Position

$$X_i^{t+1} = X_i^t + V_i^{t+1}$$

Sigmoid for Binary PSO

$$S(V) = \frac{1}{1 + e^{-V}}$$

Binary position

$$X = \begin{cases} 1, & r < S(V) \\ 0, & r \geq S(V) \end{cases}$$

Simple fitness

$$Fitness = Accuracy$$

Fitness with feature penalty

$$Fitness = Accuracy - \lambda \frac{N_{selected}}{N_{total}}$$

The easiest way to remember the main equation is:

$$\text{New Velocity} = \text{Previous Movement} + \text{Move Toward PBest} + \text{Move Toward GBest}$$

And then:

$$\text{New Position} = \text{Old Position} + \text{New Velocity}$$

For example, if your code contains:

`</> Python`

```
w = 0.7
c1 = 1.5
c2 = 1.5
```

then:

Parameter	Meaning
$w = 0.7$	Inertia weight
$c_1 = 1.5$	Influence of particle's own best
$c_2 = 1.5$	Influence of swarm's best
r_1	Random cognitive factor
r_2	Random social factor
PBest	Particle's best solution
GBest	Swarm's best solution

Example:

1. Particle Position

Each particle represents a possible solution.

The position of particle i at iteration t is:

$$X_i(t)$$

For feature selection, a particle can represent the features:

$$X_i = [x_1, x_2, x_3, \dots, x_n]$$

For example:

$$X_i = [1, 0, 1, 1, 0, 0]$$

means:

Feature	Selection
F1	1 → Selected
F2	0 → Not selected
F3	1 → Selected
F4	1 → Selected
F5	0 → Not selected
F6	0 → Not selected

Thus, the selected features are:

$$F1, F3, F4$$

2. Velocity Formula

The main PSO equation is the **velocity update equation**:

$$V_i(t + 1) = wV_i(t) + c_1r_1(PBest_i - X_i(t)) + c_2r_2(GBest - X_i(t))$$

where:

- $V_i(t + 1)$ = new velocity
- $V_i(t)$ = current velocity
- w = inertia weight
- c_1 = cognitive coefficient
- c_2 = social coefficient
- r_1, r_2 = random numbers between 0 and 1
- $PBest_i$ = particle's personal best position
- $GBest$ = global best position
- $X_i(t)$ = current position

This equation determines how the particle moves through the search space.

3. Position Update Formula

After calculating the new velocity, we update the particle's position:

$$X_i(t + 1) = X_i(t) + V_i(t + 1)$$

So the process is:

```
Current Position
  ↓
Calculate Velocity
  ↓
New Velocity
  ↓
Update Position
  ↓
New Position
```

4. Meaning of Inertia Weight w

The inertia term is:

$$wV_i(t)$$

It controls how much the particle continues moving in its previous direction.

For example:

$$w = 0.7$$

means the previous velocity has a significant influence on the new velocity.

Generally:

- Large $w \rightarrow$ more exploration
- Small $w \rightarrow$ more exploitation

So:

$w \rightarrow$ controls exploration vs. exploitation

5. Cognitive Component

The cognitive component is:

$$c_1 r_1 (PBest_i - X_i)$$

It represents the particle's own experience.

In simple language:

"I remember the best solution I have found, so I should move toward it."

This is why c_1 is called the **cognitive coefficient**.

6. Social Component

The social component is:

$$c_2 r_2 (GBest - X_i)$$

It represents the influence of the **whole swarm**.

In simple language:

"The swarm has found a good solution, so I should move toward it."

This is why c_2 is called the **social coefficient**.

7. The Three Components of Velocity

The velocity equation can therefore be understood as:

$$\text{New Velocity} = \text{Inertia} + \text{Personal Experience} + \text{Swarm Experience}$$

or:

$$V_{new} = \underbrace{wV}_{\text{Inertia}} + \underbrace{c_1 r_1 (PBest - X)}_{\text{Cognitive}} + \underbrace{c_2 r_2 (GBest - X)}_{\text{Social}}$$

This is the core PSO equation.

8. Personal Best (PBest)

Each particle remembers the best solution it has found.

Suppose a particle has obtained:

```
Iteration 1 → Accuracy = 85%
Iteration 2 → Accuracy = 89%
Iteration 3 → Accuracy = 87%
Iteration 4 → Accuracy = 92%
```

Then:

$$PBest = 92\%$$

The corresponding feature subset becomes the particle's personal best.

9. Global Best (GBest)

The entire swarm also remembers the best solution found by **any particle**.

Suppose:

Particle	Best Accuracy
P1	91%
P2	94%
P3	89%
P4	92%

Then:

$$GBest = 94\%$$

The feature subset associated with P2 becomes the global best.

10. Fitness Function for Feature Selection

In your problem, the fitness function evaluates how good a selected feature subset is.

A simple fitness function is:

$$Fitness = Accuracy$$

For example:

Features	Accuracy
F1,F2,F3	90%
F1,F3,F4	94%
F2,F4,F5	82%

Therefore:

$$GBest = 94\%$$

11. Fitness Function with Feature Reduction

For feature optimization, we may want high accuracy with fewer features.

One possible formula is:

$$Fitness = Accuracy - \lambda \left(\frac{N_{selected}}{N_{total}} \right)$$

where:

- $Accuracy$ = prediction accuracy
- $N_{selected}$ = number of selected features
- N_{total} = total number of features
- λ = penalty coefficient

This prevents PSO from simply selecting every feature.

12. Binary PSO for Feature Selection

Feature selection is naturally a **binary** problem:

$$x_j \in \{0, 1\}$$

where:

0 = feature not selected

1 = feature selected

Standard PSO operates in continuous space, so a **binary PSO** often uses a sigmoid function.

13. Sigmoid Formula

The sigmoid function is:

$$S(V) = \frac{1}{1 + e^{-V}}$$

It converts the velocity into a value between 0 and 1.

For example, if:

$$V = 2$$

then:

$$\begin{aligned} S(2) &= \frac{1}{1 + e^{-2}} \\ &\approx 0.881 \end{aligned}$$

So there is approximately an **88.1% probability** of selecting the feature.

If:

$$V = -2$$

then:

$$S(-2) \approx 0.119$$

14. Binary Position Update

After calculating the sigmoid:

$$S(V) = \frac{1}{1 + e^{-V}}$$

we generate a random number r :

$$r \sim U(0, 1)$$

Then:

$$X_{ij}(t+1) = \begin{cases} 1, & r < S(V_{ij}) \\ 0, & otherwise \end{cases}$$

Thus the feature is either selected or not selected.

15. Example of Binary PSO

Suppose:

$$V = 1.5$$

Then:

$$\begin{aligned} S(1.5) &= \frac{1}{1 + e^{-1.5}} \\ &\approx 0.818 \end{aligned}$$

Suppose random number:

$$r = 0.60$$

Since:

$$0.60 < 0.818$$

we select the feature:

$$X = 1$$

But if:

$$r = 0.90$$

then:

$$0.90 > 0.818$$

and:

$$X = 0$$